Lesson Plan

Part 1:

Overview

- Title of the unit: **RoBACHtic Symphony**
- Target grade: **5th Grade**
- Problem statement:
  - How do you determine a violin's bow movement, placement, and techniques (bow posture) while playing a musical piece with a variety of notes?
  - How does a computer handle sound without any resonating chamber, strings, and pipes unlike any other traditional instruments: wind, percussion and string instruments
- Learning objective
  - Computer Science
    1) CS1: Students will be able to distinguish between analog and digital and understand that digital data are much easier to process with the electronics.
    2) CS2: Students will be able to realize the given machine is a basic synthesizer.
    3) CS3: Students willl be able to develop the idea that technology could promote the aethestic activities.
  - Music
    1) M1: Students will be able to understand the importance of bow movement, position, and techniques (bow posture).
    2) M2: Students will be able to determine bow movement, position, and techniques (bow posture) when reading sheet music.
    3) M3: Students will be able to interpret bow position notation in sheet music.

Part 2: GSE Standards

- **Computer Science:**
  - ***Innovative Designer and Creator: CSS.IDC.3-5.4*** Use a variety of technologies within a design process to identify and solve problems by creating new, useful, or imaginative solutions.
    3. Modify, remix, or incorporate portions of an existing program into one's own work, to develop something new or add more advanced features.

Considering the inherent limitations of the machine, like requiring transition time between notes and playing just one string at a time, students are encouraged to develop inventive strategies to work around these restrictions, especially as the complexity of their chosen song escalates. Some viable approaches students might explore include:

1. Collaborating with traditional instruments, assigning the machine the sections that are less demanding.
2. Employing multiple machines simultaneously to circumvent the identified limitations.
3. Adapting the chosen song to accommodate the machine's specific constraints.

➢ ***Creative Communicator: CSS.CC.3-5.6*** Select and use the most appropriate platform, tool, style, format, and digital media to clearly and creatively express thoughts, messages, goals, or positions.

1. Create original works or responsibly repurpose or remix digital resources into new creations.
2. Communicate complex ideas clearly and effectively by creating or using a variety of digital objects.

Students will learn that the machine generates music by synthesizing pre-recorded violin sounds—pre-made digital resources—effectively engaging in a form of remixing. In this sense, they could express their emotions and thoughts by utilizing the digital resources composing the music.

❖ **Music**:

➢ ***Creating: ESGM5.CR.2*** Compose and arrange music within specified guidelines.

a. Create rhythmic and melodic motives to enhance literature.
b. Compose music (with or without text) within an octave scale in simple meter (e.g. quarter notes, quarter rests, barred eighth notes, half notes, half rests, dotted half notes, barred sixteenth notes, whole notes, whole rests, dotted quarter notes, single eighth notes, eighth rests, triplets).
c. Arrange rhythmic patterns to create simple forms, instrumentation, and various styles.

➢ ***Performing: ESGM5.PR.3*** Read and Notate music.

a. Read, notate, and identify, in various meters, iconic, and standard notation (e.g. quarter notes, quarter rests, barred eighth notes, half notes, half rests, dotted half notes, barred sixteenth notes, whole notes, whole rests, dotted quarter notes, single eighth notes, eighth rests, triplets).
b. Read and notate melodic patterns within a treble clef staff.
c. Read, notate, and identify standard symbols (e.g. repeat sign, bar line, double bar line, time signatures, crescendo/decrescendo, 1st and 2nd endings, coda, accent mark, accelerando/ritardando, sharp/flat).

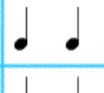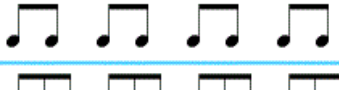
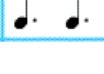
Before the performance segment, students will engage in a brief instructional session where they will acquire basic knowledge on reading sheet music and understanding fundamental principles of music composition. This preparatory phase will empower them with a grasp on the majority of the music vocabulary listed below, with particular emphasis placed on a few crucial concepts. Upon completing practice sessions with the machine using a set of simple, pre-selected songs, participants will be tasked to create a piece of music that effectively complements the functions and features of the assigned machine.

❑ Vocabulary

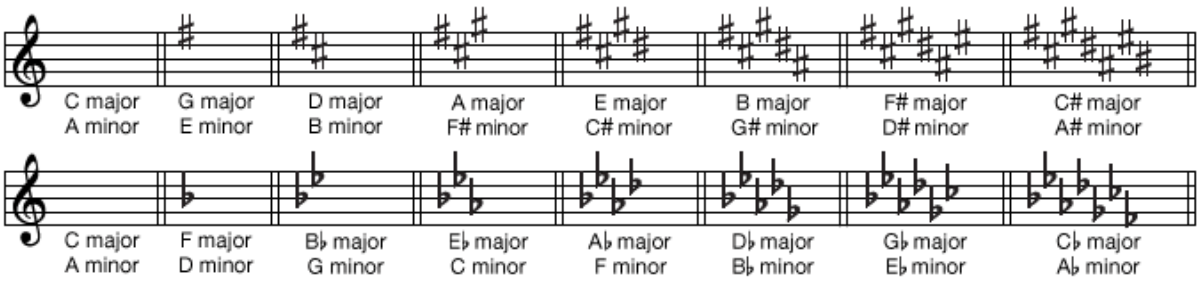
- **Computer Science:**
  - **Analog** – A defining characteristic of data; analog data are stored in a continuous transmission of a signal. It is often contrasted with digital, which is how computers store and process data as a set of individual symbols.
  - **Event** – An action or occurrence recognized by software, often originating from the external environment, that may be handled by the software.
  - **Remix** – To change a set of code by adding or rearranging smaller code segments to create a different outcome.
  - **Sequence** – An ordered, step-by-step process of an action or event proceeding in a pattern.
- **Music**:
  1. **Sheet Music Vocabulary:**
     - **Motive / Phrase / Period / Section** – units consisting a sheet music
     - **Bar line / Double bar line / Bold double bar line –** lines that separate units of a sheet music
     - **Repeat sign** – A sign that indicates a section should be repeated
     - **Note / Rest** – A music notation indicating the length of sound and silence e.g. barred notes, dotted notes, triplets

ITEM	NOTE	REST	VALUE (number of beats)
Whole note/rest			4
Half note/rest			2
Quarter note/rest			1
Eighth note/rest			1/2
Sixteenth note/rest			1/4

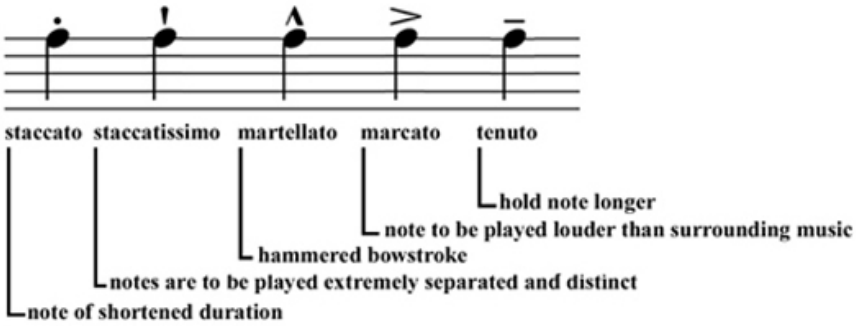
     - **Time signature** – A music notation indicating how many beats are in each measure

	time signature	beat unit	division of the beat
Simple Duple			
Compound Duple			
Simple Triple			
Compound Triple			
Simple Quadruple			
Compound Quadruple			
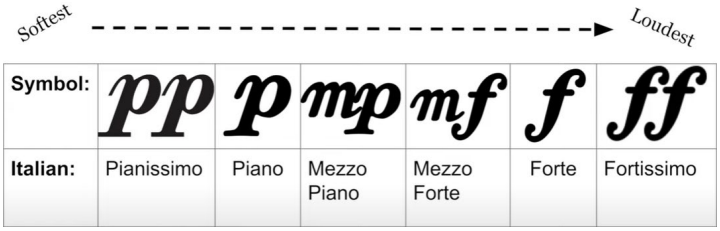
- **Key signature** - the arrangement of signs on particular lines and spaces of a musical staff to indicate that the corresponding notes, in every octave, are to be consistently raised (by sharps) or lowered (by flats) from their natural pitches.
  e.g. sharp (♯), flat (♭), natural (♮)



- **Accent** – A music notation indicating a a louder dynamic and a stronger attack to apply to a single note or an articulation mark.



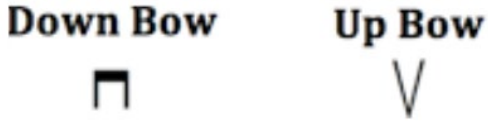
- **Dynamics** - A music notation indicating the variation in loudness between notes
  e.g. crescendo / decrescendo



- **Tempo –** A music notation for the speed at which the composer wants a piece of music performed
  e.g. accelerando / ritardando

Tempo	Definition	bpm
Largo	slowly and broadly	40-60
Adagio	slowly	66-76
Andante	at a walking pace	76-108
Moderato	moderately	108-120
Allegretto	moderately fast	112-120
Allegro	fast and bright	120-168
Vivace	lively and fast	168-176
Presto	exceptionally fast	168-200

2. **Violin Vocabulary:**
    - **Upbow / Downbow –** Direction of a bow stroke



    - **Legato / Slur –** A music notion indicating that the notes are played continously

Part 3: Lesson Plan

1. Lesson Overview (3-5 minutes):
    a. Brief introduction to the lesson and its objectives.
    b. Introduce the four-stringed Classical Orchestra instruments with a focus on the violin.
    c. Show examples of violin instrumentals in music and movie soundtracks to open their mind and ears to intentional hearing of these instruments in other musical environments
2. Understanding the Violin & Basic Music Theory (7-10 minutes):
    a. Discussion on violin strings and their corresponding open notes on sheet music.
    b. Introduction to the octave scale and essential music theory symbols and concepts: up-bow, down-bow, legato, staccato, etc.
3. Introduction to Sound Synthesis and Digital Music (7-10 minutes):
    a. Brief explanation of analog vs. digital data with a focus on sound.
    b. Introduction to sound synthesis: how digital devices like computers can generate and manipulate sounds without traditional instrument components.
    c. Explanation that the machine they will assemble is a basic synthesizer, showcasing the blend of technology with music.
4. Kit Assembly and Familiarization (10-15 minutes):
    a. Students read the instruction manual and assemble the educational kits.
    b. Explanation and identification of key components, with an emphasis on how each part contributes to sound synthesis and manipulation.
5. Practice & Debugging Session (15-20 minutes):
    a. Lead students in playing "Twinkle-Twinkle Little Star" using their kits.
    b. Support them through any debugging issues, ensuring their machines function correctly.
    c. Play a snippet of a classical piece from a famous movie or a piece that has been frequently reused and have students identify and notate the bow movements.
    d. Repeat the practice session with this piece, guiding students through the process.
6. Creative Composition & Peer Review (20-25 minutes):
    a. Instruct students to compose a short melody on a blank piece of sheet music, adhering to a rubric requiring at least five uses of bow notation, at least three unique notes, and a total of 10-20 notes.
    b. Have students play their composed pieces for a peer, who will provide constructive feedback based on the rubric.

Part 4: Pedagogical notes

Bow	By interactively observing proper bow techniques on a model violin, the student will be able to

Technique	make the connection between theoretical concepts and their application. This will strengthen their understanding and pattern recognition of bow movement, position, and techniques. **Tips for Educators:** Use visuals, demonstrations, and videos to illustrate the concepts effectively. Help train their hearing by engaging students with questions that make them distinguish different sounds based on the bow technique.
Remix	As the students write their sheet music, they will remix a musical progression. This creative process can help further develop the student's ability to determine bow technique when listening to any violin instrumentals. **Tips for Educators:** Use simple pieces to help students grasp the concept then gradually showcase more complex compositions in mainstream productions. Use demonstrations and music samples to help explain the significance of different bow positions on sound production.
Interpret bow notation in sheet music	Learning how to interpret bow notation in sheet music is critical to include a visual demonstration. By controlling the mechatronic arm when playing a sample piece of music, the students will gain a more intuitive audio-visual understanding of bow notation. **Tips for Educators:** Provide memorable sheet music examples with bowing notations (up-bow, down-bow, legato, staccato, etc.).
Understand Digital	Students should develop an understanding of how digital sound synthesis works and how it differs from acoustic sound production. They'll explore the concept of digital versus analog signals, learning how the machine they're working with synthesizes violin sounds digitally. **Tips for Educators:** Incorporate visual aids and sound samples that distinctly demonstrate the differences between digital and analog sounds. Facilitate hands-on activities where students can manipulate digital sound and immediately hear the results.